

UDAIPUR, India

WMO: 425420

Lat: 24.618N

Lon: 73.896E

Elev: 1684

StdP: 13.82

Time Zone: 5.50 (IND)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 44.8	(c) 47.4	(d) 26.3	(e) 21.6	(f) 71.9	(g) 30.0	(h) 25.7	(i) 73.8	(j) 11.5	(k) 71.6	(l) 9.8	(m) 71.2	(n) 1.3	(o) 320	(p) 0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 20.9	(c) 105.9	(d) 69.5	(e) 103.9	(f) 68.9	(g) 101.8	(h) 68.7	(i) 80.0	(j) 88.8	(k) 79.0	(l) 86.6	(m) 78.2	(n) 85.6	(o) 7.1	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	155.2	83.0	77.2	151.0	82.0	76.6	147.8	81.5	45.3	89.2	44.2	85.9	43.4	85.9	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.2	(b) 10.7	(c) 9.4	(e) 40.2	(f) 109.4	(g) 2.7	(h) 3.0	(i) 38.3	(j) 111.6	(k) 36.8	(l) 113.3	(m) 35.3	(n) 115.0	(o) 33.3	(p) 117.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.4	61.0	67.5	76.7	86.1	91.8	89.5	83.1	80.5	80.7	78.1	70.5	63.0		
	DBStd	10.37	4.12	4.90	5.41	4.35	3.58	4.42	3.40	1.95	2.52	3.20	3.71	4.16		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	256	136	28	1	0	0	0	0	0	0	0	3	88		
	CDD50	10010	342	490	828	1083	1296	1186	1026	946	922	872	615	404		
	CDD65	4791	13	98	364	633	831	736	561	481	472	407	168	27		
	CDH74	64959	286	1178	4826	9955	13986	11138	6665	4698	4999	4838	1959	431		
	CDH80	34678	34	360	2519	6343	9616	6917	2709	1247	1941	2350	593	49		
(14) Wind	WSAvg	3.5	2.6	3.0	3.5	4.2	5.4	5.6	4.8	3.9	3.2	2.2	1.9	2.1		
(15) Precipitation	PrecAvg	25.0	0.1	0.1	0.2	0.3	0.3	2.4	9.5	6.1	4.6	0.5	0.1	0.0		
	PrecMax	40.4	1.4	0.5	1.2	1.5	1.7	8.0	17.8	15.8	11.1	3.0	3.1	0.4		
	PrecMin	14.2	0.0	0.0	0.1	0.0	0.0	0.0	3.2	0.2	0.5	0.0	0.0	0.0		
	PrecStd	6.8	0.3	0.1	0.2	0.4	0.5	1.5	3.5	4.0	3.1	0.8	0.6	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	90.9	101.5	106.7	109.7	109.5	98.5	91.1	95.6	95.3	89.8	84.9		
		MCWB	61.4	62.4	65.1	65.8	68.6	74.5	74.9	76.8	73.8	68.0	63.6	62.7		
	2%	DB	80.2	87.7	97.6	103.9	106.2	105.6	94.0	88.9	93.2	93.4	87.6	81.0		
		MCWB	59.5	62.3	67.1	65.7	69.4	72.5	76.3	77.1	73.8	68.1	64.7	61.2		
	5%	DB	77.2	84.7	94.8	101.9	104.5	101.8	91.6	87.4	91.1	91.7	85.7	78.6		
		MCWB	58.1	61.3	65.6	65.6	69.0	73.5	76.4	76.7	73.6	68.1	64.1	59.9		
	10%	DB	74.1	81.4	91.9	99.9	102.6	98.4	89.6	85.9	89.0	90.1	83.7	76.4		
		MCWB	56.8	60.3	63.2	65.8	69.2	74.6	76.1	76.3	73.9	67.8	63.4	59.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	67.7	76.8	74.8	79.3	83.0	80.5	80.0	79.8	76.8	72.6	67.0		
		MCDB	78.0	81.7	95.9	95.0	95.8	98.1	87.9	86.0	86.3	85.1	80.0	78.6		
	2%	WB	61.4	63.9	71.1	70.6	76.9	80.3	79.3	79.1	78.5	75.0	70.4	63.4		
		MCDB	74.9	81.9	89.8	93.6	95.2	91.9	86.0	84.4	85.0	84.0	78.0	77.2		
	5%	WB	59.5	62.7	66.0	68.6	75.2	78.8	78.5	78.2	77.5	73.6	67.7	61.6		
		MCDB	73.5	81.1	89.5	92.8	94.3	89.6	84.9	83.3	83.7	82.9	78.3	75.0		
	10%	WB	57.9	61.3	64.3	67.4	73.8	77.8	77.8	77.6	76.7	71.9	65.8	60.0		
		MCDB	71.9	78.9	87.8	93.2	93.9	89.2	84.2	82.5	82.9	82.8	79.1	73.4		
(35) Mean Daily Temperature Range	MDBR		24.9	25.7	26.7	25.3	20.9	15.6	9.9	9.4	13.7	23.0	24.0	24.5		
	5% DB	MCDBR	29.8	29.9	29.7	27.5	22.7	20.2	13.8	12.8	19.0	25.4	27.1	27.6		
		MCWBR	12.9	11.3	10.4	7.6	5.9	6.4	3.4	4.2	5.8	7.3	9.6	11.6		
	5% WB	MCDBR	26.1	26.2	26.6	23.4	19.7	15.7	10.3	9.7	12.1	18.1	19.4	23.9		
		MCWBR	12.5	11.4	11.2	9.1	7.7	6.1	4.4	4.3	5.0	7.4	8.9	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.388	0.394	0.442	0.499	0.548	0.638	0.663	0.601	0.531	0.478	0.473	0.420		
	taud		2.187	2.198	2.067	1.960	1.884	1.742	1.718	1.864	2.002	2.031	1.978	2.086		
	Ebn at Noon		272	280	272	259	245	222	216	230	244	250	240	254		
	Edh at Noon		41	44	52	60	64	74	75	65	55	51	51	44		
(44) All-Sky Solar Radiation	RadAvg		1367	1672	2012	2260	2332	1988	1423	1346	1642	1736	1429	1295		
	RadStd		63	44	69	61	50	134	151	135	150	65	87	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page